

Mohamed Tarek Abdelhady

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate with a strong foundation in Embedded Systems, Industrial Automation, PCB Design, Internet of Things (IoT), and Renewable Energy Systems. Skilled in PLC programming using Siemens TIA Portal, classic control systems, embedded C programming, and hardware-software integration through academic projects and hands-on training.

Passionate about automation, control systems, and industrial technologies, with a commitment to continuous learning and solving real-world engineering challenges. Active member of IEEE and Google Developer Student Clubs (GDSC). Seeking a Junior Automation Engineer position where I can apply my technical knowledge, contribute to innovative automation solutions, and grow in a professional engineering environment.

TECHNICAL SKILLS

- **Automation & Control:** PLC Programming (Siemens TIA Portal, Ladder Logic), Classic Control Systems, Variable Frequency Drives (VFD), Control Systems Design, Industrial Panel Building
- **Embedded Systems:** Arduino Development, Electronic Circuit Design, PCB Design, IoT Systems
- **Software & Tools:** Siemens TIA Portal, Factory I/O, Altium Designer, Proteus, Multisim, CAdE_SIMU, EKTS, Arduino IDE, MATLAB/Simulink, Firebase
- **Programming Languages:** Python, Embedded C, Ladder Logic

EDUCATION

Bachelor of Engineering in Electronics & Communication Engineering Expected July 2026
Higher Institute of Engineering and Technology in Manzala, Egypt

- **Relevant Coursework:** MATLAB/Simulink, PCB Design, Digital Electronics, Communication Systems, Renewable Energy Systems, Control Systems, Microprocessors

PROFESSIONAL EXPERIENCE

HA (Hassan Ali Consulting Group) August 2025
Industrial Automation Trainee 140 hours

- Completed comprehensive training in Classic Control, PLC programming, Motor Drives, and Panel Building
- Programmed PLC systems using Siemens TIA Portal for various industrial applications
- Configured Variable Frequency Drives (VFDs) for speed and direction control

3M Company August 2024
Engineering Intern 160 hours

- Gained exposure to industrial engineering practices and manufacturing processes
- Assisted in quality control and process improvement initiatives
- Collaborated with engineering teams on production optimization projects

Petrobel (Belayim Petroleum Company) July 2025
Engineering Intern 56 hours

- Observed and assisted with petroleum engineering operations
- Participated in safety protocol training and compliance checks
- Contributed to documentation and reporting of operational procedures

PROJECTS

Design, Modeling, Implementation, and IoT-Based Monitoring of a 60-Watt Photovoltaic Renewable Energy System

- Designed and implemented a complete 60-W photovoltaic renewable energy system with integrated IoT monitoring capabilities
- Modeled and simulated PV system performance using MATLAB/Simulink for optimal efficiency analysis
- Implemented TL494-based buck converter for efficient battery charging regulation
- Developed conventional inverter using timer 555 for basic AC power conversion
- Developed pure sine wave inverter using EGS002 SPWM controller module for clean power output
- Created IoT monitoring board for real-time data collection and analysis using Firebase
- Developed responsive web interface for remote monitoring of system parameters

PLC Programming Projects (Siemens TIA Portal)

- Developed and tested PLC programs for industrial applications including: Color Detection System, Pick-and-Place System, Elevator Control System, Automatic Filling Tank, Traffic Light Control, Cyclic Motor Operation, and Temperature Monitoring System
- Utilized Ladder Logic programming with timers, counters, comparators, and mathematical operations
- Implemented HMI interfaces for user-friendly control and monitoring of industrial processes
- Simulated and validated programs using Factory I/O industrial simulation software

Classic Control Projects

- Designed and implemented control circuits including: Sequential Motor Control, Star-Delta Starter with Forward/Reverse, Automatic Washing Machine Controller, and Timed Motor Operations
- Created detailed wiring diagrams and tested circuit functionality under various load conditions
- Developed comprehensive documentation for each control system design

Smart Blind Glasses

- Developed wearable assistive device using ultrasonic sensors for obstacle detection and navigation assistance
- Implemented dual feedback system using haptic (vibration motor) and auditory (buzzer) signals for user awareness
- Designed portable, low-power solution optimized for visually impaired users
- Conducted user testing and feedback collection for continuous improvement

RFID-Based Attendance & Person Counter

- Designed RFID-based system for automated attendance tracking and room occupancy counting
- Implemented automated lighting control based on occupancy status for energy efficiency
- Developed user-friendly interface for administrator access and data visualization

CERTIFICATIONS & TRAINING

• Industrial Automation Training Classic Control, PLC Programming, Motor Drives, Panel Building — HA Consulting Group	140 hours, Aug 2025
• CCNA (Cisco Certified Network Associate) Networking fundamentals, routing, switching, and network security	24 hours, Jul 2024
• Python & Machine Learning Python programming, data analysis, and machine learning fundamentals	24 hours, Jun 2025
• Home Automation Training Smart home systems, IoT integration, and automation protocols	Jul–Aug 2024
• Network Fundamentals L1 Basic networking concepts and protocols	Jul 2023

LEADERSHIP & ACTIVITIES

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- **IEEE Member (Vice Chair 2025)** — Active participant in technical workshops and engineering community events; led student chapter activities and organized technical seminars; coordinated team projects and mentored junior members
 - **Google Developer Student Clubs (GDSC) Head Technical 2025** — Engaged in collaborative technology learning, development initiatives, and hackathons; contributed to open-source projects and community development